

ASSIGNMENT NO.6: Data Manipulation and Table Management

Student Name	Sandesh Dixit	Roll Number	20988
Batch	SY. BSc Data Science	Topic	Work Book(Assignment no-6)

Q1) Create a MySQL database named CollegeDB and create two tables: Course and Student.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Age	INT	
City	VARCHAR(30)	
CourseID	INT	FOREIGN KEY

QUERY:

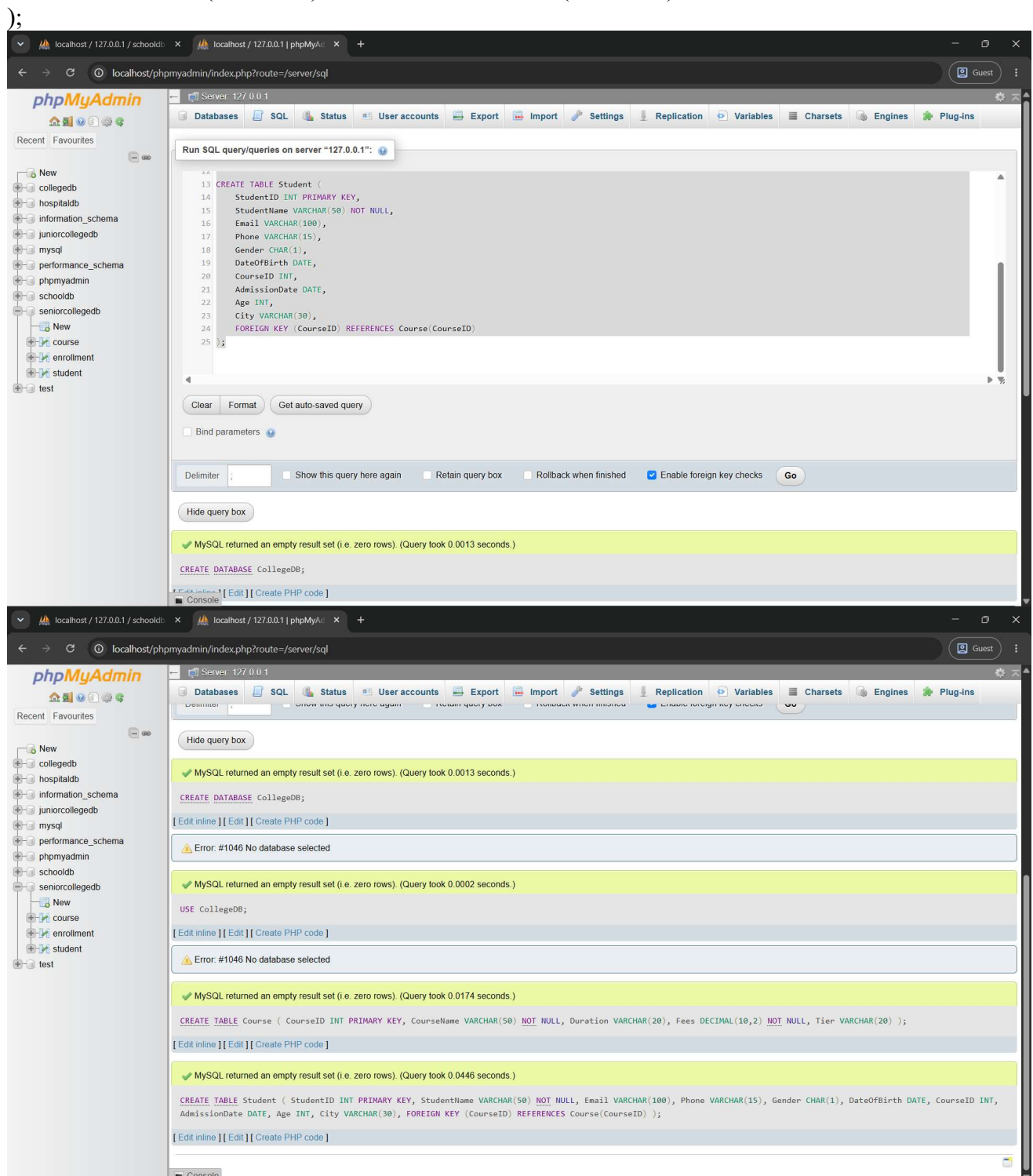
```
CREATE DATABASE CollegeDB;
```

```
USE CollegeDB;
```

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50) NOT NULL,  
    Duration VARCHAR(20),  
    Fees DECIMAL(10,2) NOT NULL,  
    Tier VARCHAR(20)  
);
```

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50) NOT NULL,  
    Email VARCHAR(100),  
    Phone VARCHAR(15),  
    Gender CHAR(1),  
    DateOfBirth DATE,
```

CourseID INT,
 AdmissionDate DATE,
 Age INT,
 City VARCHAR(30),
 FOREIGN KEY (CourseID) REFERENCES Course(CourseID)



Q2) Insert at least 4 records into the Course table and 6 records into the Student table.

Make sure every student has a valid CourseID from the Course table.

QUERY:

INSERT VLAUES:

INSERT INTO course (CourseID, CourseName, Duration, Fees, Tier) VALUES

(101, 'B.Sc IT', '3 Years', 35000.00, NULL),
(102, 'BCA', '3 Years', 40000.00, NULL),
(103, 'BBA', '3 Years', 50000.00, NULL),
(104, 'Bsc DATA SCIENCE', '3 Years', 60000.00, NULL),
(105, 'MCA', '3 Years', 45000.00, NULL),
(106, 'B.Com', '3 Years', 60000.00, NULL),
(107, 'CSE', '4 Years', 80000.00, NULL),
(108, 'MSE', '2 Years', 50000.00, NULL);

INSERT INTO student (StudentID, StudentName, Email, Phone, Gender, DateOfBirth, CourseID, AdmissionDate, Age, City) VALUES

(1, 'Amit Sharma', 'amit@gmail.com', '9876543210', 'M', '2005-04-12', 101, '2026-07-07', 20, 'Mumbai'),
(2, 'Ram Mohan', 'ram@gmail.com', '9845563210', 'M', '2007-04-16', 102, '2025-06-17', 21, 'Delhi'),
(3, 'Anisa Patil', 'anisa@gmail.com', '9848809543', 'F', '2006-05-12', 103, '2024-04-03', 22, 'Pune'),
(4, 'Rita Nayak', 'rita@gmail.com', '9267563710', 'F', '2004-08-15', 104, '2025-03-23', 20, 'Mumbai'),
(5, 'Rohan Mehta', 'abc@gmail.com', '9875643993', 'M', '2004-11-25', 105, '2026-07-08', 23, 'Delhi'),
(6, 'Sneha Joshi', 'sneha@gmail.com', '9854326710', 'F', '2009-08-23', 106, '2026-07-08', 21, 'Nashik'),
(7, 'Arjun Nair', 'arjun@gmail.com', '9876540321', 'M', '2009-08-24', 107, '2026-08-05', 24, 'Pune'),
(8, 'Aditya Ballal', 'aditya@gmail.com', '9876004532', 'M', '2009-08-25', 108, '2026-05-18', 21, 'Mumbai'),
(9, 'Vikas Sharma', 'vikas@gmail.com', '9876541111', 'M', '2005-01-10', 102, '2026-07-26', 20, 'Thane');

OUTPUT:

localhost / 127.0.0.1 / schooldb

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/sql&pos=0&db=collegedb&table=course

Server: 127.0.0.1 > Database: collegedb > Table: course

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 7 (8 total, Query took 0.0003 seconds)

SELECT * FROM `course`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

		CourseID	CourseName	Duration	Fees	Tier
☐	Edit Copy Delete	101	B.Sc IT	3 Years	35000.00	NULL
☐	Edit Copy Delete	102	BCA	3 Years	40000.00	NULL
☐	Edit Copy Delete	103	BBA	3 Years	50000.00	NULL
☐	Edit Copy Delete	104	Bsc DATA SCIENCE	3 Years	60000.00	NULL
☐	Edit Copy Delete	105	MCA	3 Years	45000.00	NULL
☐	Edit Copy Delete	106	B.Com	3 Years	60000.00	NULL
☐	Edit Copy Delete	107	CSE	4 Years	80000.00	NULL
☐	Edit Copy Delete	108	MSE	2 Years	50000.00	NULL

Check all With selected: Edit Copy Delete Export

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

localhost / 127.0.0.1 / schooldb

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/sql&pos=0&db=collegedb&table=student

Server: 127.0.0.1 > Database: collegedb > Table: student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Showing rows 0 - 8 (9 total, Query took 0.0003 seconds)

SELECT * FROM `student`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

		StudentID	StudentName	Email	Phone	Gender	DateOfBirth	CourseID	AdmissionDate	Age	City
☐	Edit Copy Delete	1	Amit Sharma	amit@gmail.com	9876543210	M	2005-04-12	101	2026-07-07	20	Mumbai
☐	Edit Copy Delete	2	Ram Mohan	ram@gmail.com	9845563210	M	2007-04-16	102	2025-06-17	21	Delhi
☐	Edit Copy Delete	3	Anisa Patil	anisa@gmail.com	9848809543	F	2006-05-12	103	2024-04-03	22	Pune
☐	Edit Copy Delete	4	Rita Nayak	rita@gmail.com	9267563710	F	2004-08-15	104	2025-03-23	20	Mumbai
☐	Edit Copy Delete	5	Rohan Mehta	abc@gmail.com	9875643993	M	2004-11-25	105	2026-07-08	23	Delhi
☐	Edit Copy Delete	6	Sneha Joshi	sneha@gmail.com	9854326710	F	2009-08-23	106	2026-07-08	21	Nashik
☐	Edit Copy Delete	7	Arjun Nair	arjun@gmail.com	9876540321	M	2009-08-24	107	2026-08-05	24	Pune
☐	Edit Copy Delete	8	Aditya Ballal	aditya@gmail.com	9876004532	M	2009-08-25	108	2026-05-18	21	Mumbai
☐	Edit Copy Delete	9	Vikas Sharma	vikas@gmail.com	9876541111	M	2005-01-10	102	2026-07-26	20	Thane

Check all With selected: Edit Copy Delete Export

Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

Q3) Write a MySQL query to insert one new student record into the Student table.

The record could include:

Field
StudentID
StudentName

Age
City
CourseID

QUERY:

--Q3) Insert one new student record

INSERT INTO Student

(StudentID, StudentName, Age, City, CourseID)

VALUES

(10, 'Rohit Desai', 21, 'Mumbai', 103);

-- Verify: SELECT * FROM Student

WHERE StudentID =10;

OUTPUT:

The screenshot shows the phpMyAdmin web interface. The left sidebar displays a database structure tree with 'college' selected. The main panel shows the 'Run SQL query/queries on database college:' section. The SQL query entered is:

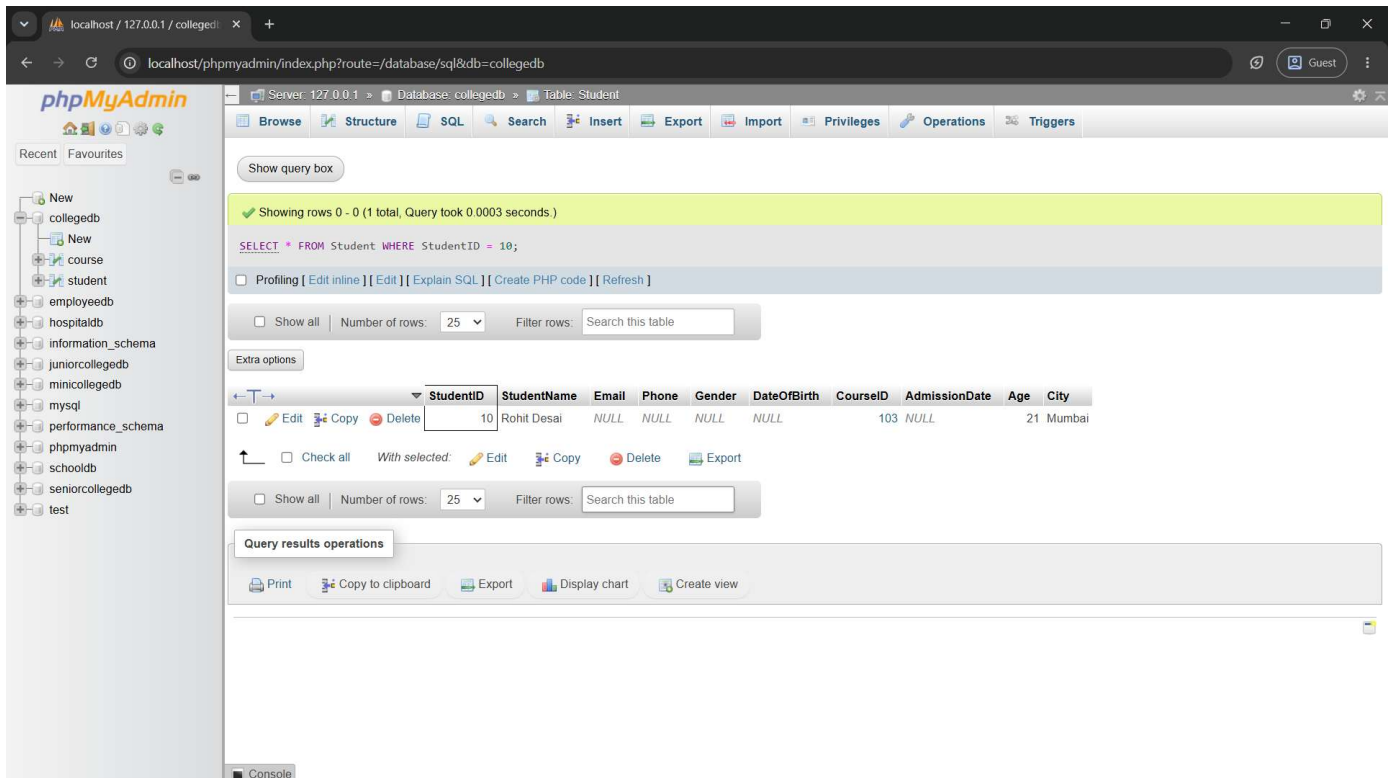
```

1 -- Q3) Insert one new student record
2
3 INSERT INTO Student
4 (StudentID, StudentName, Age, City, CourseID)
5 VALUES
6 (10, 'Rohit Desai', 21, 'Mumbai', 103);
7
8 -- Verify:
9 SELECT * FROM Student
10 WHERE StudentID =10;

```

Below the query editor, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'. A checkbox for 'Bind parameters' is present. At the bottom, there are options for 'Delimiter' (set to semicolon), 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (checked). A 'Go' button is at the bottom right.

The execution result is displayed in a green bar at the bottom: **✓ 1 row inserted. (Query took 0.0277 seconds.)**



Q4) Write a MySQL query to update the city of a student using StudentID.

Example requirement:

Change Required
Update the city of StudentID 2 to Mumbai

QUERY:

-- Q4) Update the city of StudentID 2 to Mumbai

UPDATE Student

SET City = 'Mumbai'

WHERE StudentID = 2;

-- Verify:

SELECT * FROM Student

WHERE StudentID = 2;

OUTPUT:

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

- New
- collegedb
 - New
 - course
 - student
- employeeedb
- hospitaldb
- information_schema
- juniorcollegedb
- minicollegedb
- mysql
- performance_schema
- phpmyadmin
- schooldb
- seniorcollegedb
- test

Server: 127.0.0.1 > Database: collegedb > Table: Student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on database collegedb:

```
1 UPDATE Student
2 SET City = 'Mumbai'
3 WHERE StudentID = 2;
4 -- Verify:
5 SELECT * FROM Student
6 WHERE StudentID = 2;
7
8
```

Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Simulate query Go

Hide query box

1 row affected. (Query took 0.0087 seconds)

UPDATE Student SET City = 'Mumbai' WHERE StudentID = 2;

[Edit inline] [Edit] [Create PHP code]

Console

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

- New
- collegedb
 - New
 - course
 - student
- employeeedb
- hospitaldb
- information_schema
- juniorcollegedb
- minicollegedb
- mysql
- performance_schema
- phpmyadmin
- schooldb
- seniorcollegedb
- test

Server: 127.0.0.1 > Database: collegedb > Table: Student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Simulate query Go

Hide query box

1 row affected. (Query took 0.0087 seconds)

UPDATE Student SET City = 'Mumbai' WHERE StudentID = 2;

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 0 (1 total, Query took 0.0003 seconds)

-- Verify: SELECT * FROM Student WHERE StudentID = 2;

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

	StudentID	StudentName	Email	Phone	Gender	DateOfBirth	CourseID	AdmissionDate	Age	City
☐	2	Ram Mohan	ram@gmail.com	9845563210	M	2007-04-16	102	2025-06-17	21	Mumbai

☐ Check all With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Console

Q5) Write a MySQL query to update the fees of a course using CourseID.

Example requirement:

Change Required
Update the fees of CourseID 101 to 47000

QUERY:

Q5) Update fees of CourseID 101 to 47000

UPDATE Course

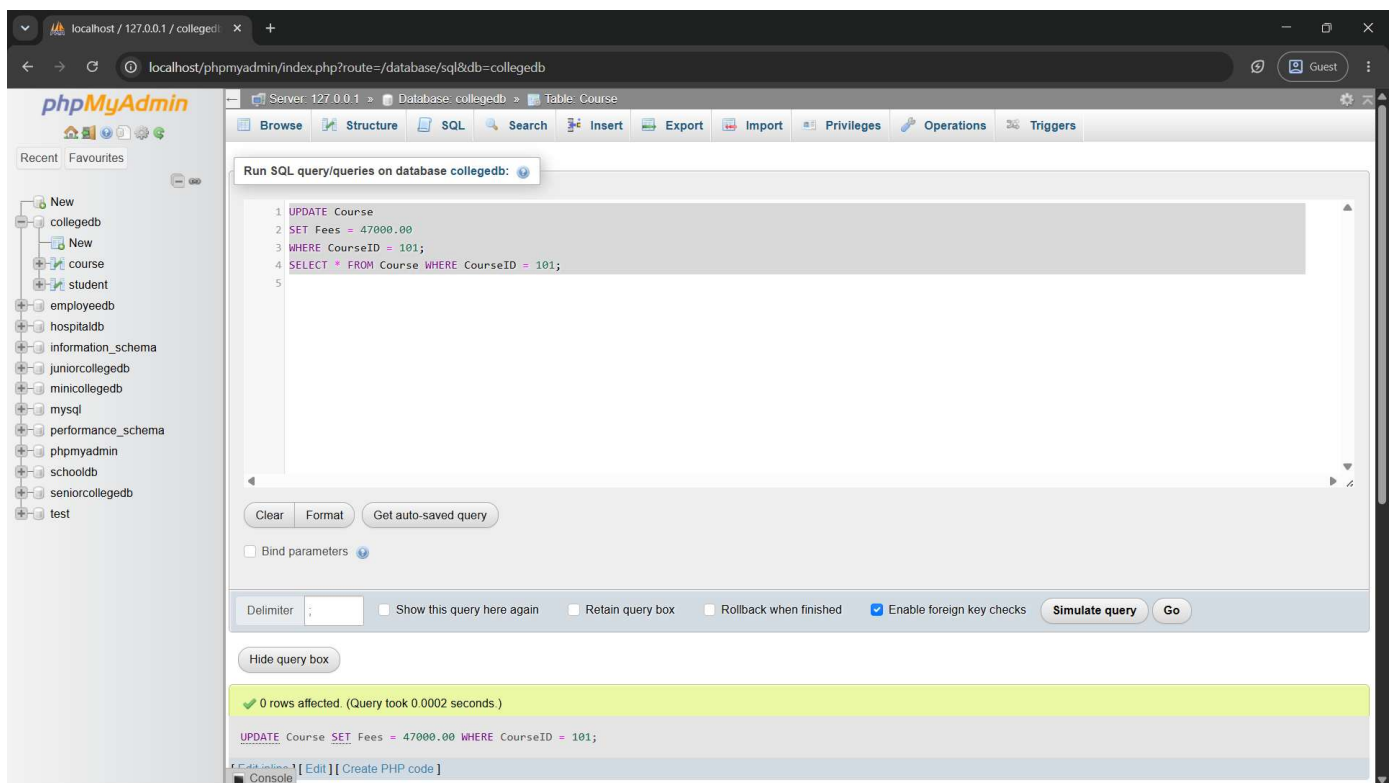
SET Fees = 47000.00

WHERE CourseID = 101;

Verify:

SELECT * FROM Course WHERE CourseID = 101;

OUTPUT:

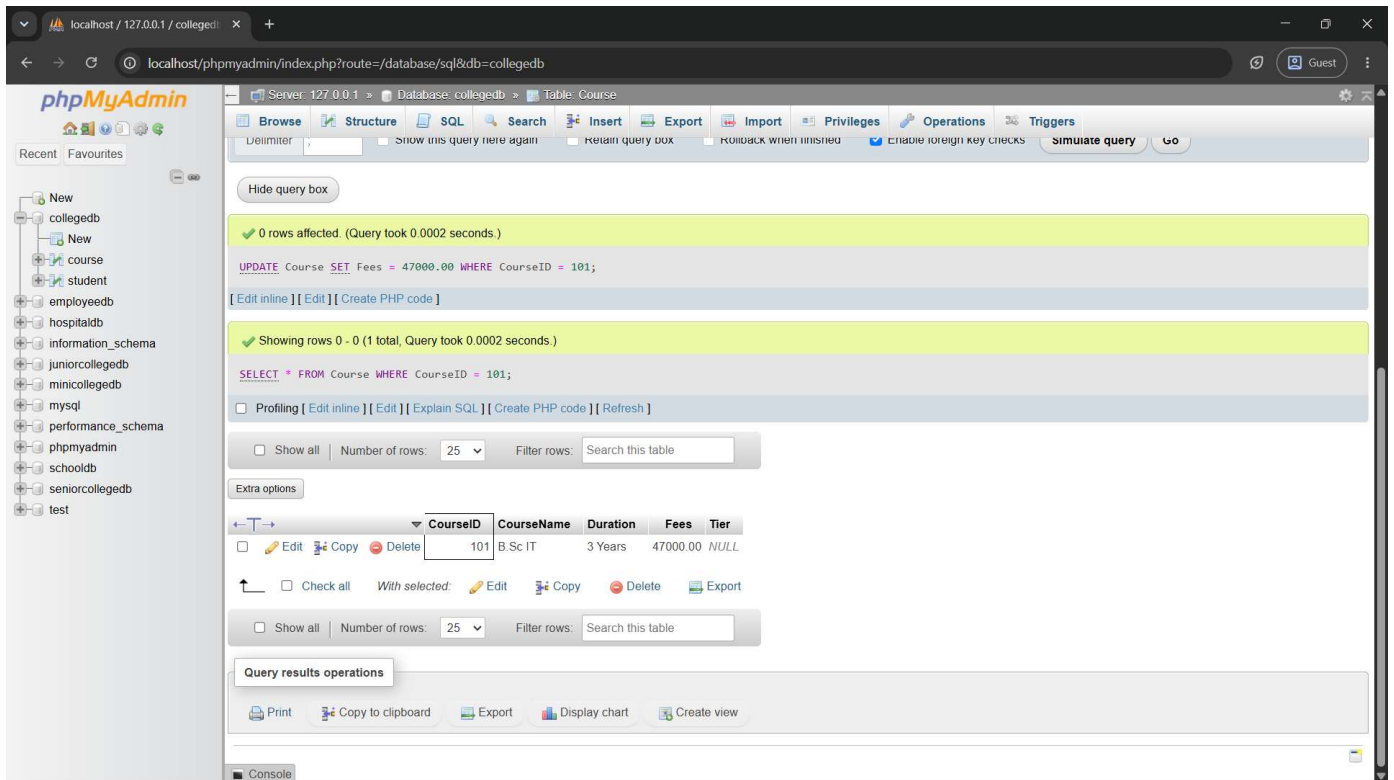


The screenshot shows the phpMyAdmin web interface in a browser window. The address bar indicates the URL is localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb. The interface is for the 'collegedb' database, specifically the 'Course' table. The SQL query editor contains the following query:

```
1 UPDATE Course
2 SET Fees = 47000.00
3 WHERE CourseID = 101;
4 SELECT * FROM Course WHERE CourseID = 101;
5
```

Below the query editor, there are buttons for 'Clear', 'Format', and 'Get auto-saved query'. There is also a checkbox for 'Bind parameters'. At the bottom of the query editor, there are checkboxes for 'Show this query here again', 'Retain query box', 'Rollback when finished', and 'Enable foreign key checks' (which is checked). There are also 'Simulate query' and 'Go' buttons.

The execution result is displayed in a green bar, indicating '0 rows affected (Query took 0.0002 seconds)'. Below this, the executed query is shown: 'UPDATE Course SET Fees = 47000.00 WHERE CourseID = 101;'. At the bottom, there are links for 'Edit this query' and 'Create PHP code', and a 'Console' tab is visible.



Q6) Write a MySQL query to delete one student record using StudentID.

Also write one line explaining why the WHERE clause is important while using DELETE.

DELETE FROM Student

WHERE StudentID = 6;

Verify:

SELECT * FROM Student;

The WHERE clause ensures that only the specified student record is deleted.

Without WHERE, all records from the Student table would be deleted.

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

- New
- collegedb
 - New
 - course
 - student
- employeeedb
- hospitaldb
- information_schema
- juniorcollegedb
- minicollegedb
- mysql
- performance_schema
- phpmyadmin
- schooldb
- seniorcollegedb
- test

Server: 127.0.0.1 > Database: collegedb > Table: Student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Run SQL query/queries on database collegedb:

```
1 DELETE FROM Student
2 WHERE StudentID = 6;
3
4 -- Verify:
5 SELECT * FROM Student;
6
```

Clear Format Get auto-saved query

☐ Bind parameters

Delimiter ; ☐ Show this query here again ☐ Retain query box ☐ Rollback when finished ☒ Enable foreign key checks Simulate query Go

Hide query box

✓ 1 row deleted. (Query took 0.0043 seconds)

DELETE FROM Student WHERE StudentID = 6;

[Edit inline] [Edit] [Create PHP code]

localhost / 127.0.0.1 / collegedb

localhost/phpmyadmin/index.php?route=/database/sql&db=collegedb

phpMyAdmin

Recent Favourites

- New
- collegedb
 - New
 - course
 - student
- employeeedb
- hospitaldb
- information_schema
- juniorcollegedb
- minicollegedb
- mysql
- performance_schema
- phpmyadmin
- schooldb
- seniorcollegedb
- test

Server: 127.0.0.1 > Database: collegedb > Table: Student

Browse Structure SQL Search Insert Export Import Privileges Operations Triggers

Hide query box

✓ 1 row deleted. (Query took 0.0043 seconds)

DELETE FROM Student WHERE StudentID = 6;

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 8 (9 total, Query took 0.0004 seconds)

-- Verify: SELECT * FROM Student;

[Edit inline] [Edit] [Create PHP code]

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	StudentID	StudentName	Email	Phone	Gender	DateOfBirth	CourseID	AdmissionDate	Age	City			
☐	Edit	Copy	Delete	1	Amit Sharma	amit@gmail.com	9876543210	M	2005-04-12	101	2026-07-07	20	Mumbai
☐	Edit	Copy	Delete	2	Ram Mohan	ram@gmail.com	9845563210	M	2007-04-16	102	2025-06-17	21	Mumbai
☐	Edit	Copy	Delete	3	Anisa Patil	anisa@gmail.com	9848809543	F	2006-05-12	103	2024-04-03	22	Pune
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☐	Edit	Copy	Delete	8	Aditya Ballal	aditya@gmail.com	9876004532	M	2009-08-25	108	2026-05-18	21	Mumbai
☐	Edit	Copy	Delete	9	Vikas Sharma	vikas@gmail.com	9876541111	M	2005-01-10	102	2026-07-26	20	Thane
☐	Edit	Copy	Delete	10	Rohit Desai	NULL	NULL	NULL	NULL	103	NULL	21	Mumbai

Console Check all With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)